

permissions for that file will be:

```
-rw-r--r--
```

```
owner          Read & Write (rw-)
group  Read (r--)
others  Read (r--)
```

This is very similar to the permissions that your `/etc/shadow` file (which contains all local user passwords) will have:

```
user@host:/home/user# ls -l /etc/shadow
```

```
-rw-r----- 1 root shadow 869 2010-06-14 18:27 /etc/
shadow
```

```
user@host:/home/user#
```

Permissions:

```
owner = Read & Write (rw-)
```

```
group = Read (r--)
```

```
other = None (---)
```

Ownership:

```
owner = root
```

```
group = shadow
```

Now that you are familiar with permissions and understand what the `ls` output means, let us get on with changing permissions. The command that we will be using here is `chmod`. A very important thing here is to be cautious while playing around with permissions, because you do not want to create security flaws on your system, or give unrestricted access to unauthorised personnel.

`Chmod` can be used with letters, as well as numbers, which should be pretty obvious to you if you understood the permission tables given above. While using the letters, the various options in the `chmod {options} filename` command that you have are in the accompanying table.

Example:

We have 4 files, named A, B, C and D, with read and write permission for the owner and read for the rest.

Now if we want to add owner execute bit to A, the command will be:

```
user@host:/home/user$ chmod
u+x A
```

```
user@host:/home/user$ ls -l A
-rwxr--r-- 1 user user 0 Jun
14 16:14 A
```

If we want to add other write & execute bit

Options	Definition
u	owner
g	group
o	other
x	execute
w	write
r	read
+	add permission
-	remove permission
=	set permission